Subject: COMPILER DESIGN	**Type:** Revision Notes
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Unit 1: Lexical Analysis Formulas & Lex Tool

1. Role of the Lexical Analyzer

The lexical analyzer reads the input source program character-by-character, groups them into lexemes, and produces a sequence of tokens. It also strips out whitespaces and comments while keeping track of line numbers for error reporting.

2. Regular Expressions to Finite Automata

Thompson's Construction converts Regular Expressions into Non-deterministic Finite Automata (NFA). Subset Construction algorithm converts NFA into equivalent DFA. Hopcroft's algorithm minimizes DFA states.

3. LEX Specification Structure

A LEX file consists of three sections separated by '%%': Definitions (macros & header inclusions), Rules (patterns with corresponding C actions), and User Subroutines (main function and auxiliary logic).

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